

TSETSEN UUL, Mongolia

WMO: 442240

Lat: 48.748N

Lon: 96.004E

Elev: 2181

StdP: 77.72

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.1	-42.0	-46.8	0.0	-44.8	-44.1	0.1	-41.8	7.7	-12.4	7.2	-13.9	0.1	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	25.0	13.4	23.2	12.6	21.6	11.9	15.0	21.9	13.9	20.6	12.9	19.7	2.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	11.9	17.8	11.2	10.8	16.5	10.1	10.0	15.9	50.0	21.8	46.3	20.5	43.2	19.7	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	7.0	6.3	-46.4	28.1	3.9	1.7	-49.3	29.3	-51.6	30.4	-53.8	31.3	-56.6	32.6		
(5)			-46.3	16.1	4.2	0.6	-49.3	16.6	-51.8	17.0	-54.2	17.3	-57.2	17.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.4	-28.9	-25.1	-14.7	-0.9	5.2	11.8	14.1	11.6	5.1	-3.5	-15.4	-24.8	(6)
(7)		DBStd	16.16	7.01	7.32	8.47	6.37	4.52	3.67	3.01	3.56	4.41	5.99	7.40	6.79	(7)
(8)		HDD10.0	5908	1205	982	764	328	159	24	3	25	156	419	763	1080	(8)
(9)		HDD18.3	8652	1463	1215	1022	578	408	196	133	211	397	678	1013	1338	(9)
(10)		CDD10.0	302	0	0	0	1	9	78	132	73	9	0	0	0	(10)
(11)		CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0	(11)
(12)		CDH23.3	144	0	0	0	0	1	27	82	33	1	0	0	0	(12)
(13)		CDH26.7	13	0	0	0	0	0	1	11	2	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	0.9	1.2	1.9	2.6	2.9	2.2	2.0	1.9	2.2	2.1	2.0	1.2	(14)
(15)	Precipitation	PrecAvg	221	3	3	5	8	16	32	62	52	19	7	7	5	(15)
(16)		PrecMax	332	13	9	13	22	42	60	137	95	59	18	14	12	(16)
(17)		PrecMin	140	0	0	0	1	2	7	10	12	3	1	2	0	(17)
(18)		PrecStd	50	3	2	3	5	11	13	33	21	13	5	3	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.3	-0.3	6.7	16.1	22.0	25.7	28.4	26.7	21.8	12.8	6.8	-2.5	(19)
(20)			MCWB	-7.7	-3.6	0.6	7.3	10.3	13.4	16.2	13.6	11.6	4.9	1.6	-6.0	(20)
(21)		2%	DB	-10.0	-5.4	4.0	13.5	19.2	23.9	25.6	23.9	18.8	10.3	3.5	-5.8	(21)
(22)			MCWB	-11.7	-8.2	-0.8	6.1	9.2	12.4	13.8	12.6	10.0	3.7	-0.8	-8.1	(22)
(23)		5%	DB	-12.5	-8.9	1.6	11.3	17.1	22.0	23.6	21.8	16.6	8.4	-0.1	-9.6	(23)
(24)			MCWB	-13.9	-10.7	-2.3	4.4	7.9	11.6	13.2	12.0	8.7	2.4	-3.2	-11.3	(24)
(25)		10%	DB	-15.9	-11.9	-0.6	9.1	14.9	20.3	21.9	19.8	14.5	6.4	-3.4	-12.4	(25)
(26)			MCWB	-16.9	-13.4	-3.8	3.0	6.5	10.9	12.6	11.2	7.4	1.2	-5.8	-13.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.7	-3.7	1.1	7.9	11.7	15.1	18.0	15.6	12.8	5.7	2.0	-5.8	(27)
(28)			MCDWB	-5.4	-0.5	6.1	15.2	20.2	23.3	25.4	23.1	19.4	11.4	6.5	-2.9	(28)
(29)		2%	WB	-11.7	-8.1	-0.7	6.2	9.8	13.7	15.9	14.1	10.6	4.3	0.3	-8.0	(29)
(30)			MCDWB	-10.0	-5.6	3.6	13.0	18.1	21.5	22.2	20.8	17.9	10.1	4.2	-5.7	(30)
(31)		5%	WB	-13.9	-10.8	-2.1	4.7	8.3	12.6	14.5	12.9	9.0	3.2	-2.0	-11.1	(31)
(32)			MCDWB	-12.5	-8.8	1.6	10.9	16.1	19.7	20.6	19.9	15.6	8.5	1.2	-9.6	(32)
(33)		10%	WB	-16.9	-13.3	-3.8	3.1	7.0	11.6	13.4	11.9	7.9	1.8	-4.5	-13.5	(33)
(34)			MCDWB	-15.8	-11.9	-0.6	8.6	14.2	18.6	19.9	18.5	13.6	6.7	-2.1	-12.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.8	16.5	17.4	12.3	13.0	13.2	12.0	12.6	13.7	13.8	13.1	12.6	(35)
(36)			MCDBR	16.7	18.1	16.4	14.5	17.5	16.3	16.2	16.6	17.4	16.3	14.6	14.6	(36)
(37)			MCWBR	15.3	16.0	12.5	8.7	10.0	8.0	7.6	8.2	10.1	9.9	7.1	12.9	(37)
(38)		5% WB	MCDWB	16.9	18.0	15.9	13.8	15.7	14.2	13.0	14.1	16.0	15.5	10.2	14.6	(38)
(39)			MCWBR	15.4	16.0	12.3	8.7	9.6	7.6	7.0	7.5	9.6	10.1	6.9	13.0	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.213	0.237	0.275	0.314	0.316	0.318	0.320	0.303	0.270	0.255	0.235	0.217	(41)	
(42)		taud	2.343	2.290	2.258	2.296	2.332	2.409	2.437	2.457	2.504	2.472	2.393	2.326	(42)	
(43)		Ebn at Noon	893	937	944	935	944	941	934	938	946	908	852	834	(43)	
		Edh at Noon	65	89	110	119	120	112	108	100	86	74	62	58		
(44)	All-Sky Solar Radiation	RadAvg	1.58	2.66	4.25	5.64	6.67	6.81	6.41	5.60	4.61	3.00	1.62	1.21	(44)	
(45)		RadStd	0.12	0.14	0.15	0.25	0.35	0.24	0.34	0.25	0.21	0.17	0.15	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	-1.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page